

SUNIL KULALI

Data Engineer

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PROFESSIONAL SUMMARY

Results-driven **Data Engineer** with hands-on experience in building **ETL/ELT pipelines, SQL development, data modelling**, and BI dashboard automation. Proficient in Python, SQL, Apache Airflow, Spark (PySpark), Kafka, PostgreSQL, BigQuery, and cloud platforms (AWS, GCP). Experienced in end-to-end pipeline development, data warehouse design (star/snowflake schema), and real-time data processing. Strong foundation in data quality, pipeline monitoring, and stakeholder-ready reporting. Passionate about building scalable, reliable data infrastructure that powers data-driven decision-making.

TECHNICAL SKILLS

Programming Languages	Python (Pandas, NumPy, PySpark), SQL, Bash / Shell Scripting, R (Basic)
Data Pipeline & ETL	Apache Airflow, Kafka, Spark (PySpark), Flink (Basic), Fivetran, Airbyte, dbt
Databases & Warehouses	PostgreSQL, MySQL, SQL Server, BigQuery, Snowflake, Redshift — CTEs, Window Functions, Query Optimisation, Stored Procedures
Cloud Platforms	AWS (S3, Glue, Lambda, Redshift, EC2), GCP (BigQuery, Cloud Storage, Dataflow), Azure (Basic)
Data Formats & Protocols	Parquet, Avro, ORC, JSON, CSV, REST APIs, Change Data Capture (CDC)
DevOps & Infrastructure	Docker, Git, GitHub, GitHub Actions (CI/CD), Terraform (Basic), Linux / Bash
Data Modelling	Star Schema, Snowflake Schema, Data Vault, SCD Types, OLAP vs OLTP
Monitoring & Quality	Data validation, pipeline observability, Great Expectations (Basic), MLflow (Basic)
BI & Visualisation	Power BI (DAX, Power Query), Tableau, Streamlit, Plotly, Seaborn
Productivity & Reporting	Excel (Pivot Tables, Power Query), Jira, Confluence, SharePoint, Teams

INTERNSHIP EXPERIENCE

Data Analyst Intern | IT Vedant | Bengaluru

June 2024 – April 2025

- Designed and maintained **ETL workflows** using Python (Pandas, NumPy) and SQL to ingest, transform, and load 10,000+ record datasets into structured data stores, enabling reliable KPI reporting for stakeholders.
- Automated **end-to-end reporting pipelines** using Power BI and Power Query, reducing manual reporting effort by **40%** and improving data accuracy and pipeline integrity.
- Developed **complex SQL queries** including CTEs, window functions, and stored procedures to perform data segmentation, trend analysis, and performance reporting across multiple business units.
- Built and maintained **process documentation and SOP-aligned data validation workflows** to ensure pipeline stability and data quality for downstream consumption.

Machine Learning Intern | CODE-B Solutions Pvt. Ltd. | Mumbai

March 2025 – May 2025

- Designed and optimised **data preprocessing pipelines** using Python and SQL, handling outlier treatment, feature engineering, and data cleaning — reducing model training errors by **~15%** and ensuring high data quality across datasets.
- Built **complex SQL queries, views, and stored procedures** to extract, transform, and analyse sales, customer behaviour, and performance trend data with full documentation.
- Developed structured **end-to-end data pipelines** from raw ingestion to model-ready feature sets, applying supervised ML models (Random Forest, XGBoost, Logistic Regression) on real-world business datasets.

PROJECTS

1. Real-Time Retail Sales Data Pipeline & BI Dashboard

Tech Stack: Python, PostgreSQL, REST API, Google Sheets API, Apache Airflow, Power BI, DAX, Google Apps Script

- **Built a real-time data pipeline** integrating a web data-entry portal, a serverless REST API, Google Sheets as a lightweight data store, and an Apache Airflow DAG for scheduled ingestion into PostgreSQL.
- **Designed DAX measures and star-schema data models** in Power BI for live KPI tracking across revenue, product performance, and sales trends; self-refreshing dashboard with automated report generation.
- Implemented **duplicate detection logic and data validation checks** in the pipeline to ensure data integrity end-to-end.

2. AI-Powered Data Intelligence & ETL Automation Platform

Tech Stack: Python, LangChain, LLMs, PostgreSQL, FastAPI, Power BI, RAG, Streamlit, dbt

- **Architected an end-to-end analytics pipeline** integrating LLMs and RAG to enable natural-language querying of structured PostgreSQL data, reducing ad hoc reporting time by **60%**.
- Built a **LangChain-based agent pipeline** that auto-generates and executes SQL queries from user prompts, renders results in Streamlit, and auto-produces executive-ready insight narratives.
- **Designed automated BI workflows** using Power BI (DAX, Power Query) and FastAPI REST endpoints for real-time KPI tracking across sales, revenue, and customer metrics.

3. Loan Approval Data Pipeline & Credit Risk Analytics

Tech Stack: Python, Pandas, NumPy, Scikit-learn, PostgreSQL, SQL, Streamlit, Joblib

- **Engineered an end-to-end ML data pipeline** on 45,000 loan records — from raw ingestion and data cleaning to feature engineering, model training (Random Forest: **90.2% accuracy, 0.9665 ROC-AUC**), and deployment via Streamlit.
- Designed **5 domain-driven features** (debt-to-income ratio, income-per-age, loan-to-credit-history ratio, high-loan flag, high-income flag), with SQL-based data segmentation powering downstream model inputs.
- Implemented **robust preprocessing pipelines** with systematic outlier treatment, scaling, and encoding — production-grade data quality for ML consumption.

4. Customer Churn ETL Pipeline & Retention Analytics Dashboard

Tech Stack: Python, Scikit-learn, SQL, PostgreSQL, Power BI, EDA, Pandas

- **Built a SQL + Python ETL pipeline** to ingest, clean, and segment customer data; identified key churn drivers through SQL-based cohort analysis and Python EDA.
- **Developed churn probability scoring models** and segmented at-risk customers; visualised results in Power BI dashboards enabling targeted retention strategies for marketing teams.

EDUCATION

Bachelor of Engineering – Computer Science & Engineering

2020 – 2024

Visvesvaraya Technological University | CGPA: 7.82 / 10

CERTIFICATIONS

- Data Analytics & Visualisation Job Simulation – Accenture
- Machine Learning with Python – IBM
- Data Analysis with Python – IBM
- Databases & SQL for Data Science with Python – IBM
- Python for Data Science – IBM

KEY COMPETENCIES

ETL / ELT Pipeline Development • Data Modelling (Star / Snowflake / Data Vault) • SQL Development & Optimisation • Apache Airflow • Apache Spark / PySpark • Apache Kafka • Cloud Data Engineering (AWS, GCP) • Data Warehousing (BigQuery, Snowflake, Redshift) • Data Quality & Validation • CI/CD for Pipelines • Docker & Containerisation • dbt Transformations • Real-Time Streaming • Data Governance • BI & Dashboard Automation • Stakeholder Reporting